Math 54 — Week 4

Solutions

P1. The two products are different:

$$AB = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}.$$

The transposes agree:

$$(AB)^T = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad B^T A^T = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix},$$

as the rule $(AB)^T = B^T A^T$ says.

P2.

$$X^2 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}, \qquad X^3 = X^2 X = \begin{bmatrix} 1 & -6 \\ 0 & 1 \end{bmatrix}.$$

X is a horizontal shear by -2, and doing it k times shears by $-2k$, so the guess is $X^k = \left[\begin{smallmatrix} 1 & -2k \\ 0 & 1 \end{smallmatrix}\right]$. Induction proves it: if the formula holds for k, then

$$X^{k+1} = X^k X = \begin{bmatrix} 1 & -2k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2(k+1) \\ 0 & 1 \end{bmatrix}.$$

P3.

Idea. One inverse serves every right-hand side.

$$A^{-1} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}.$$

First system:

$$A^{-1}(3,7) = \left(-9 + 14,\ \frac{15}{2} - \frac{21}{2}\right) = (5, -3).$$

Second: $A^{-1}(1,0) = (-3,\ 5/2)$, the first column of A^{-1}. Check: $3(-3) + 4(5/2) = 1$, $5(-3) + 6(5/2) = 0$.

P4.

Idea. Row reduce $[\,A \;\; I\,]$; a leading 1 in row 2 is convenient, so swap first.

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 1 & -2 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 + R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 1 & -2 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right] \xrightarrow[R_1 - R_3]{R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & 3 & -1 \\ 0 & -1 & 0 & 3 & -6 & 2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{-R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & 3 & -1 \\ 0 & 1 & 0 & -3 & 6 & -2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right] \xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -3 & 1 \\ 0 & 1 & 0 & -3 & 6 & -2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 2 & -3 & 1 \\ -3 & 6 & -2 \\ 1 & -2 & 1 \end{bmatrix}.$$

Check: row 1 of A times the columns of A^{-1}: $4 - 3 = 1$, $-6 + 6 = 0$, $2 - 2 = 0$.

P5. No. Row 2 is twice row 1, so $R_2 - 2R_1$ gives a zero row: fewer than 3 pivots, so by the invertible matrix theorem A is not invertible. (Equally, column 2 is -2 times column 1, so the columns are dependent.)

P6.

Idea. Column 2 has a single nonzero entry; after that, use the column of the minor with the most zeros.

1. Expand down column 2: only $a_{32} = 7$ is nonzero, with sign $(-1)^{3+2} = -1$. So

$$\det A = -7 \det \begin{bmatrix} 3 & 4 & 1 \\ 2 & -1 & 0 \\ 1 & 2 & 0 \end{bmatrix}.$$

2. Expand the 3×3 down its column 3: only the top entry 1 is nonzero, sign $(-1)^{1+3} = +1$:

$$\det = 1 \cdot \det \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = 4 + 1 = 5.$$

3. $\det A = -7 \cdot 5 = -35$.

P7.

1. (a) True: a nontrivial solution needs a free variable, i.e. a column without a pivot.
2. (b) True for square matrices: (h)⇒(e) in the invertible matrix theorem.
3. (c) True: (k) gives A invertible, and then $B = A^{-1}AB = A^{-1}$, which is invertible.
4. (d) True: $(A^2)^{-1} = A^{-1}A^{-1}$, by the rule for inverses of products.

P8.

Idea. Multiply out and use $A^3 = 0$; for square matrices one side is enough.

Then

$$(I - A)(I + A + A^2) = I + A + A^2 - A - A^2 - A^3 = I - A^3 = I.$$

By the invertible matrix theorem ((k)⇒(a)), $I - A$ is invertible and its inverse is $I + A + A^2$. (Compare $\frac{1}{1-a} = 1 + a + a^2 + \ldots$ for numbers.)